

**Xlore U: An Intelligent school-matching platform for Senior high school
students of Taguig City**

**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Global City**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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<Date of Proposal Defense>

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: Xlore U: An Intelligent school-matching platform
for Senior high school students of Taguig City

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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APPROVAL SHEET

This capstone project proposal titled **Xlore U: An Intelligent school-matching platform for Senior high school students of Taguig City**, prepared and submitted by **Neo Aragon V. De Villena, Gian Albert T. Orbase, Mark Justine Tibor, and Criszouie Guile P. Tolentino**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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INTRODUCTION

Project Context

The college program selection process is one of the most crucial decisions faced by graduating senior high school students, influencing their respective academic paths and career opportunities. As the K-12 program emphasizes, it aims to prepare graduates for college, employment, or career path (Parojenog et al., 2022). While career development is a lifelong process, high school students face the pivotal decision of choosing a path aligned with their interests and abilities. This also applies to Taguig City, where a multitude of graduates transition to the tertiary level without the necessary knowledge of the program they intend to pursue or the institutions that align with their interests, financial capacity, and career goals. The National Career Assessment Examination (NCAE) demonstrated a substantial correlation with the alignment of students' selected college courses. (Joel, 2025). Often leading to a program mismatch, as well as wasted time, effort, and resources.

The Philippine Statistics Authority, recorded a 4.7% increase in the economy of the city, showing continuous urban and economic expansion (“City of Taguig’s Economy Grows by 4.7 Percent in 2024,” 2025). Due to the diverse selection of programs and institutions, Overwhelms students and their parents by the magnitude of information provided. However, many students still experience difficulties in recognizing their interests, talents, and appropriate career choices, thus requiring effective career guidance (Smith & Jones, 2021). Although institutions provide guidance through orientations and admission details, these may be limited in substance and insufficient in helping students evaluate their options. Media such as digital platforms, mobile applications, educational videos, and interactive media can assist students in understanding career options and improve their readiness in making career decisions (Brown & Wilson, 2024).

Research examining the use of media as an aid in career guidance services, specifically in senior high schools, is still scarce and has yet to provide a comprehensive overview regarding the types of media used, their effectiveness, and the challenges faced in their

implementation (Lee, 2021; Anderson & Chen, 2022). The inception of Xlore U aims to address the gap by providing a centralized platform that aids senior high school graduates of Taguig City to form educated academic decisions.

Through the project, graduates are able to efficiently filter academic options, compare tuition fees, and acquire direct access to official institutional portals. Going beyond traditional data aggregation, Xlore U incorporates interactive decision-support features that aid students plot their skills against market demands in Taguig's fast paced economy. Through consolidation of contrasting information into a distinct, intuitive interface, The project aims to minimize decision making fatigue, relieve financial anxiety of parents, and ultimately empower students to forge their own academic paths that are not only ambitious but strategically aligned and lasting success.

Purpose and Description

Xlore U, serves as a centralized platform that assists senior high school graduates of Taguig City in making educated academic decisions. As many students experience the struggles of deciding which program or institution to pursue because of scattered information across various sources. Xlore U addresses this concern by centralizing essential details such as program offerings, estimated tuition fees, and direct institution links to one platform.

The platform also includes a self-assessment tool that would allow students to answer a short set of questions about their strengths and preferences. Based on those responses, the platform would suggest programs that fit their preferences. Through this platform, students can review and compare various institutions and programs while still considering their strengths and preferences. Xlore U helps reduce stress, save time, and empower students to create knowledgeable decisions about their futures.

Through bridging the gap between secondary education and the tertiary level, the platform operates as a digital guide mitigating the risks of course shifting and academic burnout. Ultimately, fostering prepared and confident students in Taguig City ensures

that every graduate is equipped with the necessary information to turn their potential into a viable career trajectory.

Objectives

The primary objective of Xlore U, is to create a centralized platform that would assist senior high school students in Taguig City in their college selection process based on their strengths, preferences and financial capabilities.

- To develop an integrative assessment tool that evaluates students' skills, interests, and preferences through quizzes, that serve as the baseline for personalized recommendations.
- To develop a filter and match feature to align with students' skills and preferences with suitable institutions and programs.
- To develop a platform that centralizes information to compare institutions and programs, and access relevant school information, along with a direct connection of official application links, to provide effective student guidance.

Scope and Limitations

The scope of the project entails the developmental phase of Xlore U, a centralized platform that aims to assist senior high school graduates in Taguig City in exploring college programs and tertiary institutions. The range of this study entails the developmental phases of Xlore U, a centralized platform aimed to guide senior high school graduates in Taguig City to explore their options regarding college programs and tertiary institutions. A platform that includes a self-assessment tool that gathers information about students' strengths and preferences. Empowering users compare and contrast institutions and their programs, estimated tuition fees, and direct links to official websites.

The project is limited to selected institutions in Taguig City and is reliant on the manual collection of data; hence, the updates regarding tuition fees, programs, and admission requirements may not be immediately reflected. Self assessment responses only serve as recommendations. The platform only serves as a research tool that students

can utilize; factors such as personal, financial, and familial considerations are beyond its scope.

Features includes:

- A platform that centralizes the information about tertiary institutions in Taguig City.
- Personalized assessments and recommendations based on students, interests, skills, and preferences.
- A platform that uses map integration to locate tertiary institutions.
- A platform that filters specialization, tuition's, and accreditation's of tertiary institutions.
- A platform that compare tertiary institutions in Taguig City.

Review of Related Literature/Studies/Systems

The development of Xlore U as a centralized platform for assisting senior high school students in Taguig City with their college selection process necessitates a thorough examination of existing systems with similar objectives. This section reviews local and foreign systems that provide college discovery, career guidance, and institution matching services. The review aims to identify the features, strengths, and limitations of these systems to establish the context for Xlore U's development and to highlight its unique contributions to the field.

According to the study of Khor and K (2023), personalized learning is an educational approach that modulates the speed, content as well as instruction in accordance with the specific needs, interests, and ability of the students to suit the demands. The strengths of this strategy are that it has the ability to address the different needs of students and enhance their interaction and performance. Individualized learning fosters a more profound understanding of the subjects, increases motivation, and improves student achievement by considering and accommodating differences among students. It may motivate students to be in control of their education and study on their own. Personalized learning can also help in eliminating educational disparity as it provides students with specialized support that may be necessary or be having some problems. Personalized learning benefits students because it accommodates their personal needs and nature in assignments, tests and learning. This is unlike the traditional one-size-fits-all method of education.

Supported by the findings of Iqbal et al. (2025) stating that the most average education system follows rules of the game, the way information is passed on to and how to evaluate it for all the students. One –size fits all strategy does not take care of individual differences such as individual differences in learner styles, cognitive abilities, motivation factors and so the total effectiveness of the teaching procedure is not satisfying. Their shortcomings are acknowledged by educators as well as the researchers; more and more they are demanding dynamic learning personalization, who are those systems of individual specific enough day to make an apparent to a particular student, his special needs and preference.

The study of Serutla et al. (2024) concurs that the e-learning role and the technology that comes with it can now support the whole assessment procedure beginning with the assignment creation, tracking of progress, marking, results communication to a point of even statistical analysis. In this respect learning analytics have become an immense mass of research enabling high end analytics and dashboards. The implementation of technology in education is regarded as a part of teaching-learning environment that can meet the needs of the learners in the 21st century. There are various styles of learning among the students yet they are largely internet-savvy and apply technology to assist them in constructing their knowledge. Students are able to refer to the internet trying to find answers. They are able to use physical materials like books to assist them. They are able to ask other people who may be more knowledgeable than them.

The study of Serutla et al. (2024) also acknowledges that Personalised learning requires data-based understanding to be put into practice. The education stakeholders have realised the radical transformation of the contemporary economy and the workforce and the necessity to equip the students with the skills and competencies more effectively to be ready to work in the workplace. The changes that mirror the system-wide shifts required to educate and train the complex skills and competencies, including creativity, innovations, problem-solving, systems thinking, digital literacies, resilience and agility. The intrinsic motivation, interests as well as cognitive capabilities and learning speed and ability of each learner vary and each learner possesses their own motivations and capabilities of self regulation. One of the most crucial missing ingredients that can help to characterise the diversity of learners and introduce personalised learning on scale is the creation of learner profiles. A learner profile is a holistic depiction of the comprehension, competencies, skills and attributes of a learner.

Insights of Kessels et al. (2024) states learning will be less complex, less boring and study success will be enhanced when student motivation is facilitated. Increased student motivation can also guarantee more intensive learning as well as deter school dropouts. Which presupposes the fact that learning and learning environment should be able to satisfy some psychological needs which would not interfere with intrinsically motivated behavior. These are the psychological needs competence, relatedness and

autonomy. One of the potential ways to solve the motivation issue is in the promotion of student autonomy in the learning activities and tests. Autonomy is said to be a significant antecedent of intrinsic motivation. A lot of teachers and schools are seeking methods of assisting personalized learning, a type of learning that enables personalized selections of learning and assessment. In personalized learning, it might be important to have flexible assessments so that learning and assessment can be more aligned. Flexible assessments can make students more motivated as they encourage student autonomy. There is however no research on flexible assessment, its implications on assessment motivation in students in a vocational education and training (VET) environment. The learning process of students will be guided by assessments and students will study knowing they are going to be assessed.

Du Plooy et al. (2024) also shares the same sentiment that learning will be less complex, less boring and study success will be enhanced when student motivation is facilitated. Increased student motivation can also guarantee more intensive learning as well as deter school dropouts. Which presupposes the fact that learning and learning environment should be able to satisfy some psychological needs which would not interfere with intrinsically motivated behavior. These are the psychological needs competence, relatedness and autonomy. One of the potential ways to solve the motivation issue is in the promotion of student autonomy in the learning activities and tests. Autonomy is said to be a significant antecedent of intrinsic motivation. A lot of teachers and schools are seeking methods of assisting personalized learning, a type of learning that enables personalized selections of learning and assessment. In personalized learning, it might be important to have flexible assessments so that learning and assessment can be more aligned. Flexible assessments can make students more motivated as they encourage student autonomy. There is however no research on flexible assessment, its implications on assessment motivation in students in a vocational education and training (VET) environment. The learning process of students will be guided by assessments and students will study knowing they are going to be assessed.

Gonzales et al. (2024) have analyzed that the most optimal education and training would be selected when individuals, particularly those in the high school level know the career they would like to follow. Students are able to choose their own career with K-12

programs. The first step that a student should take in his career is to choose a career in high school. This is a very crucial stage as it is the foundation of the student preparation. The decision to follow a certain career path should be considered thoroughly since it will influence the future and the development of a person. Today, an individual has to not only plan his career, but also think about future employment. Equally, career choice decisions are affected by student interest in the career choice of an individual. It means that senior high school students require coaching on their tracks and strands to continue at a tertiary level where they should move on after the secondary school education. Moreover, there are many factors that may influence the choice of a student in choosing a college course. One of the commendable features of SHS curriculum is the preparation of students to the professional future and challenges. Nonetheless, there exists a unique scenario whereby the tracks or strands of the students in SHS are not linked to their university course enrollments. A parallel research on the correspondence of tracks and courses in college revealed that the career decisions undertaken by the respondents have no links with their aptitude. This can lower the productivity and discontent in their respective fields. The feeling of unfulfillment and inability to achieve success can be experienced as well.

The study of Lynn and Emanuel (2021) identified that the amount of courses in the context of smart education nowadays has grown exponentially; therefore, the issue of course selection is taking a prominent part in the process of contemporary learning. Other researchers have discovered that thousands of students across the whole world have to grapple with the problem of making their most appropriate University/college courses. The Recommender Systems could be used to assist students in choosing courses that they can and are interested in. These are time saving as they offer students with expert advice based on their knowledge base component. Knowing the needs of the student can help to expand the knowledge of the student, get experience and have a greater chance of being successful. The paper is devoted to the recommender systems in the educational field that are designed to help students who are unable to make the right choice when it comes to course choice.

Algarni and Sheldon (2023) states that an efficient and effective software solution to manage the advice process effectively and efficiently is required to support the

educational process and to take off the players in the educational institutions. A course recommender system can serve as a strategic collaborator in the practice of helping the student (and the adviser) meet their educational objectives and reinforced and motivated in their study plan. But in contrast to most other existing recommendation systems, course recommender systems (CRS) have to operate on a large decision space that grows combinatorically with the number of courses; programs; and the different backgrounds, skills and goals of a student but is also limited by a variety of constraints (e.g., maximum credit hour load, course prerequisites, sequencing etc.).

The process of decision making is important in the academic life of the students. The choice that the student makes regarding the course to major in in his/her university studies, can prove or break his future. Nevertheless, there are a number of reasons both individual and course reasons that can make students decide on the course they desire to undertake when they are joining institutions of higher learning. According to (Lynn and Emanuel 2021), the factors are Future career goal that most of students select as it is the field they want to pursue. Ease, other students can take a degree simply because the degree is not challenging and they can easily pass their grades without the stress of doing so. Personally, people will find themselves in courses that are related to something that they are interested in learning. Social status, other students will choose the course that their friends are taking, so that they can do assignments together and then share ideas. The examination procedure in university, not all students are fond of majors, the final mark of which depends on their performance in written exams, and this is why they decide to take courses in which assignments and classroom activities are included in the final mark. Timetable schedule, some students opt and fail to opt courses due to the timetable. The behaviour of lecturers, some students might select courses due to the good lecturers in the department. Course format, number of course units or the number of lectures in a week and organizational course features are also minded by some students.

Based on the study by Johora et al. (2022) modern development of nations is highly dependent on the level of higher education and there is no way to preserve the level of higher education without new technologies. As part of their research they suggested a Data Driver Decision Making System, which will interpret the data of the former proposed system and will generate the results on the basis of the data provided by

more than one university, where students will be in a position to make an effective selection of the university/major on the basis of information. The analysis of the graph revealed to them that they can make the right decisions without any hassle based on the situation of admission trends, acceptance ratio, etc. when they compare the situation in the universities.

Shah et al. (2024a) have identified that there are a few systems online which the community is utilizing in order to locate appropriate universities/colleges. The Canadian colleges and schools have been using the School Finder system since 1995. Using the School Finder system, users can discover the graduate/Professional and undergraduate programs that are provided by the Canadian school and colleges. Whenever anybody wishes to locate or seek information regarding the program then he will type in the search box the requested program and it will display a user/person of all the Schools/Colleges where the Programs/Education subjects are offered at all levels in the graduate, undergraduate and secondary level. And yet there is no idea of visualizing and comparing these programs to each other and which school or college is the best among all rank-wise. In this system a person/user can use various parameters and in College Finder, one may also search about the school/college according to his/her religious concept where he has more interest. College Finder offers a search in their large database and locates the colleges/universities where a person/user has their religious inclinations as well.

Related Systems:

FindU: Find The Perfect School

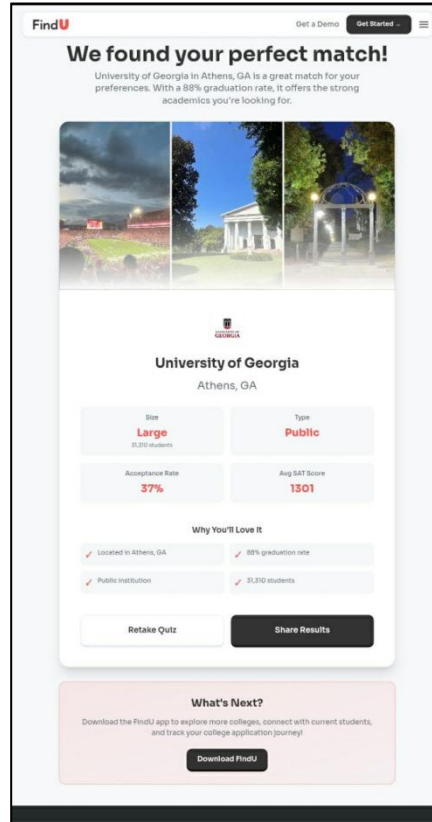


Figure 2: FindU matching interface

FindU is a mobile application developed by FindU, LLC, designed to streamline the college discovery process for students. The application allows users to swipe through college profiles, displaying key statistics such as tuition fees, financial aid availability, and test score requirements. A distinctive feature of FindU is its match score functionality, which provides users with an estimate of how well each institution aligns with their background and interests. Users can save preferred schools, build personalized lists, and compare saved institutions to identify standout options. Future developments planned for the platform include scholarship discovery tools, a discover page featuring student reviews and advice, and messaging capabilities for direct communication with admissions teams. However, as a relatively new platform, its database may not yet be comprehensive, and its focus remains primarily on institutions within the United States.

CollegeWollege

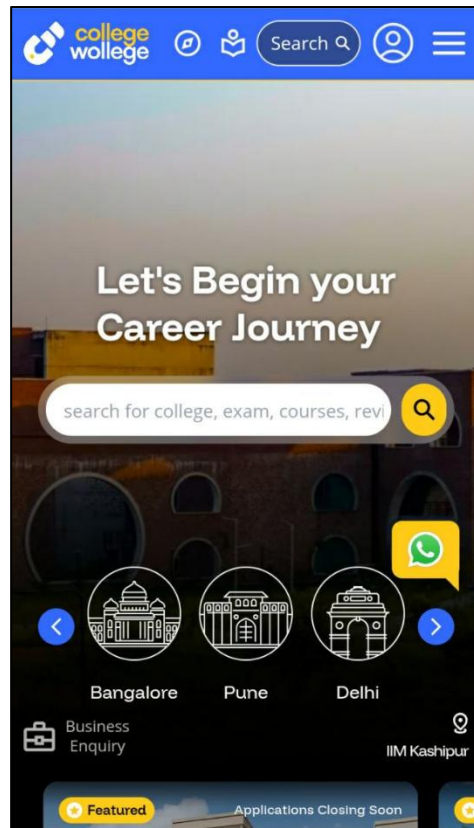


Figure 2.1: CollegeWollege landing page

Ani (2025b) analyzed that CollegeWollege is an AI-native, data-driven platform serving the Indian higher education market. The platform distinguishes itself through its emphasis on information authentication, featuring "Reviewed by" and "Fact-Checked" labels to ensure data reliability—a response to concerns about unverified content on competing platforms. CollegeWollege maintains an extensive database encompassing over 500 entrance examinations, 1,300 courses and certificates, and more than 16,000 universities and colleges. The platform offers smart search and comparison tools, personalized recommendations based on user profiles, and gamification elements designed to engage the Gen Alpha generation in the college discovery process. Additional features include virtual campus tours, video testimonials, and direct news updates from institutions through a dedicated college-side panel.

College Navigator

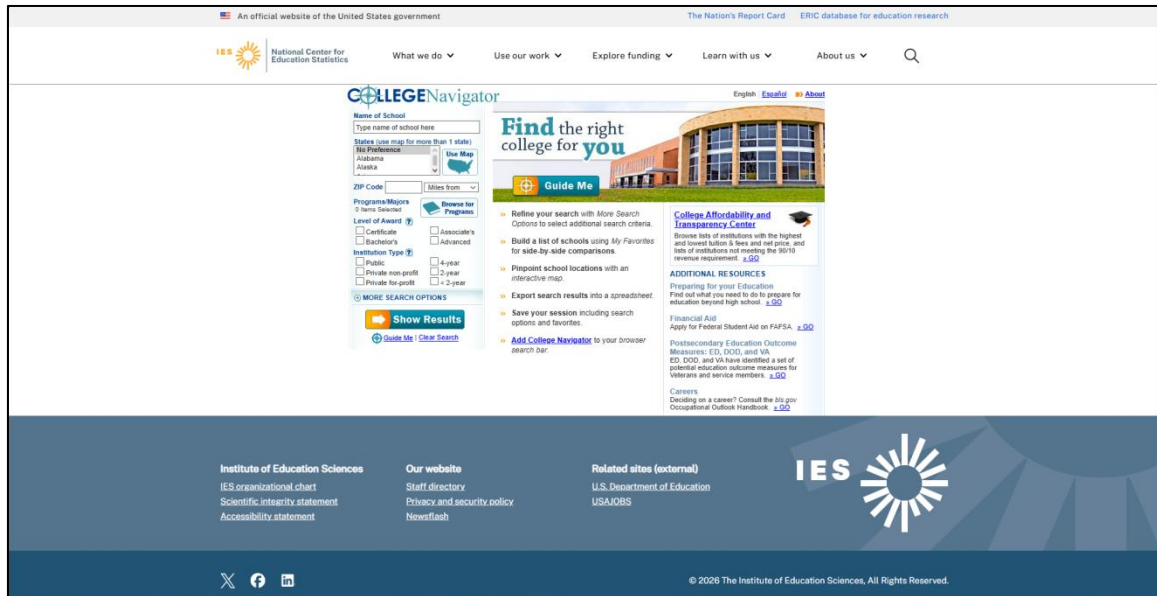


Figure 2.2: College Navigator user interface

The Applerouth Team (2024), College Navigator operated by the National Center for Education Statistics, provides what many experts consider the most robust combinable filters of any university search tool. Users can search for institutions based on specific criteria including undergraduate enrollment size, academic majors, geographic location, and SAT score ranges. The platform generates lists quickly and offers access to Common Data Sets containing detailed statistical information about each institution, including acceptance rates by applicant type, enrolled student profiles, and waitlist outcomes. Despite its powerful filtering capabilities, the platform offers limited information presentation and a less intuitive interface compared to commercial alternatives.

Coursefinder.ph

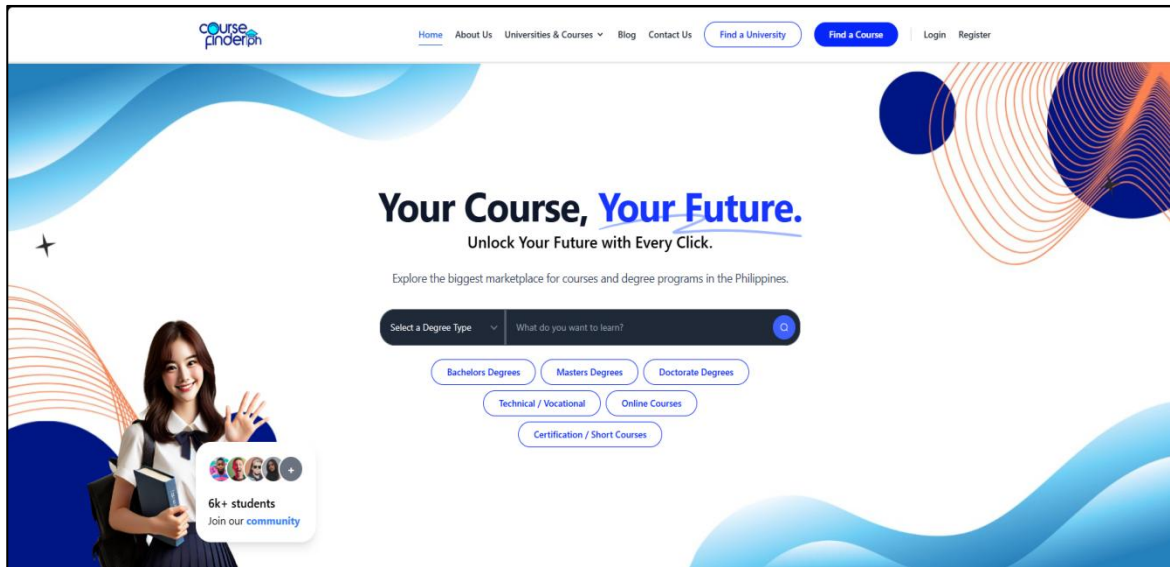


Figure 2.3: Coursefinder.ph user interface

Coursefinder.ph is an advocacy startup aimed to aid Filipinos around the globe find information on courses they want to study. Their end-goal is to create an impact on higher education through empowering course providers and Filipinos who want to further their careers and academic goals. Users can search for institutions based on courses, degree type, term structure, location, and tuition fee. The platform can also compare institutions based on said criteria enabling students to create educated decisions on their academic endeavors.

Synthesis:

In conclusion, these related studies and existing systems emphasize the significant shift in education towards personalization, data-driven decision-making, and technology based guidance. Supported by the studies of Khor and K (2023) and Iqbal et al. (2025) prove that the traditional “one-size-fits-all” approach is no longer feasible, due to their failure to address the individual differences in learning styles, interests, and cognitive abilities. Whereas, personalized learning with the assistance of adaptive technologies and learner profiling enhances engagement, motivation, and academic performance through the alignment of the educational experiences with the unique needs of each student.

Accompanying this sentiment, Serutla et al. (2024) and Du Plooy et al. (2024) have mentioned the paramount importance of technology and data analytics in enabling personalized systems. Since these technologies support unceasing evaluation, create student profiles, and support adaptive learning trajectories. Not only do they enhance the learning outcomes of students but also equip them to meet the requirements of the contemporary workforce, by developing important skills, such as critical thinking, resilience, and digital literacy. Motivation and autonomy also emerge as key factors in student success. Kessels et al. (2024) emphasizes that flexible and personalized assessment methods can enhance the inherent motivation by empowering students’ sense of control over their learning. Reinforcing the importance of systems as not merely an information provider rather empowering students to engage in meaningful decision-making processes.

In terms of career and course selection, the studies by Gonzales and Digo (2024) and Lynn and Emanuel (2021) highlight the complexity and lasting impact of these decisions. Countless students struggle with aligning their interests, abilities, and career goals, often resulting in mismatch. Recommendation and decision-support systems are essential in leading students toward educated and responsible choices, taking into account both personal preferences and institutional factors.

Existing platforms such as FindU, CollegeWollege, and College Navigator exemplify the practical application of the aforementioned concepts. Personalized matching, vast

databases, and complex filtering tools are among the capabilities offered by these systems. However, they do have significant drawbacks, such as a limited regional focus, a lack of extensive visualization and comparison capabilities, and low user engagement in some interfaces. Furthermore, many systems fail to adequately integrate localized data or address the unique needs of students in various educational situations.

In conclusion, the literature presented illustrates a disparity between the potential of tailored, data-driven educational systems and the current constraints of existing platforms. This disparity emphasizes the need for a more integrated, user-centered solution that combines personalized recommendations, detailed data, and contextual relevance. In response, Xlore U wants to address these inadequacies by creating a centralized platform dedicated to senior high school students in Taguig City that provides more relevant, accessible, and data-driven assistance with college and career decisions.

Benchmark

System Feature	FindU	College Wollege	College Navigator	Coursefinder.ph	Xlore U
Recommendation	✓	✓	✗	✓	✓
Personalization	✓	✓	✗	✓	✓
Information Completeness	✓	✓	✓	✓	✓
Navigation & Usability	✓	✓	✗	✓	✓
Accessibility	✓	✓	✓	✓	✓
User Interaction	✓	✓	✗	✓	✓
Compatibility	✓	✓	✓	✓	✓
Local Relevance	✗	✗	✗	✓	✓
Location-based Mapping	✗	✗	✗	✗	✓

Table 2: Related Systems Bench-marking

The table takes into comparison the various tertiary program research platforms such as Xlore U, Find U, College Wollege, and College Navigator. The evaluation takes into account the crucial aspects such as recommendation, personalization, information completeness, navigation & usability, accessibility, user interaction, compatibility, local relevance, and location-based mapping. Xlore U, stands out as the most feasible platform providing detailed analysis and personalized recommendations essential for effective tertiary program selection. The table also provides detailed comparisons on how each platform meets various criteria, along highlighting their respective strengths and limitations.

METHODOLOGY

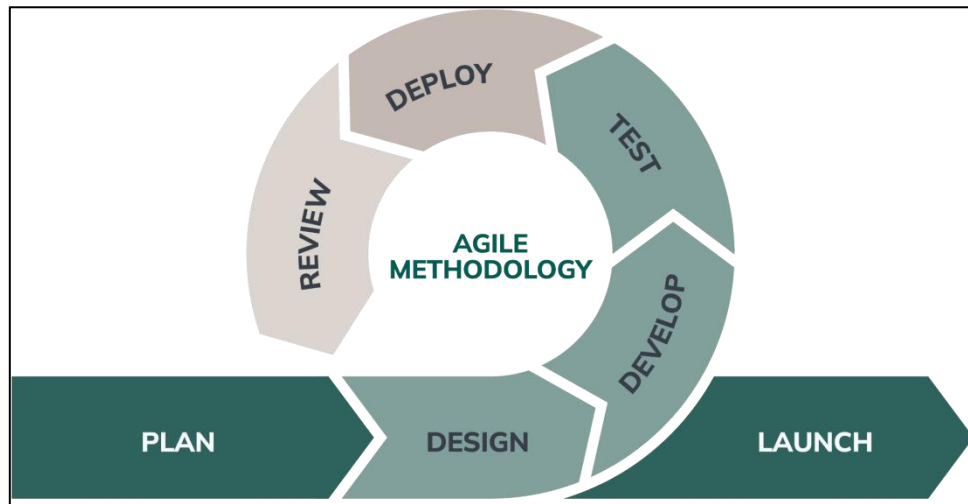


Figure 3: Agile Development Methodology

Throughout the development of Xlore U, the Agile methodology was utilized to grasp the iterative framework and inherent flexibility. The system involves numerous features including program recommendations, filtration system, and a user profile management, requiring constant improvement and testing.

The initial phase would be planning, where developers would brainstorm probable features that may benefit the users. Followed by the designing phase, where developers think of probable ways to integrate the feature. Accompanied with the development and testing phase, where features are integrated to the existing system. Succeeded by the deployment and review phase, in which the perspective features are tested upon users and given the necessary feedback. To be followed by the launching phase.

Through Agile Development, the development process is divided into smaller iterations focusing on developing specific features of the system. Through this approach the developers can adapt to the changing user requirements and feedback. Resulting in efficiency and user-centered platform.

- **Technologies to be Used**

For the inception of Xlore U, the proponents have concluded the various technologies to be utilized. The following technologies ensure that the development of the project is seamless and provide an interactive experience for users.

Xlore U is a web based platform that aids students in their academic decision-making process. It is required for users to have access to web browsers such as: Google Chrome and Microsoft Edge installed into their devices. For the development of the platform, the proponents will utilize Node.js as its server side language, as it is known for its popularity and wide range of features. For the frontend development, the proponents will utilize a combination of JavaScript, HTML, CSS known for their reliability in creating an interactive experience for users. In terms of database, the proponents will utilize MySQL, known for its straightforward integration and support for multitude of languages, allowing efficient management of storage, retrieval, and management of structured data required by the system. The language to be primarily utilized is JavaScript, to be utilized for both the frontend and backend of the platform, the platform will be primarily developed and tested using Visual Studio Code.

- **Calendar of Activities**

Month	January				February				March				April			
Activity																
Planning																
Brainstorming																
Title Proposal																
Chapter 1																
Project Context																
Objectives																
Purpose and Description																
Scope and Limitations																
Chapter 2																
Review of Related Literatures and Systems																
Synthesis																
Chapter 3																
Technical Background																
Requirement Analysis																
Requirements Documentation																
Design of System																

In Progress

Completed

Table 3: Xlore U, Calendar of Activities

The Gantt chart illustrates the activities in overview during the development of Xlore U: An Intelligent school-matching platform for Senior high school students of Taguig City. Progress has been tracked using a color coded scheme: yellow indicating in progress and blue indicating completion.

- **Resources**

Hardware

Type	Specification	Purpose
GPU	Vega3	Animations, and visual effects, offloading tasks from the CPU to ensure smooth, high-performance user experiences.
	Intel Iris Xe	
	Intel(R) Iris(R) Xe	
	Ryzen 5 7520U	
CPU	AMD Athlon 200ge 2 cores 4 thread	Processing data and executing instructions to manage all system operations.
	Intel Core i7-12700H	
	11th Gen Intel(R) Core(TM) i5-1135G	
	AMD Ryzen 5 7520U Radeon	
Memory	24 GB DDR4 2400MHZ	Used by browsers, servers, and code editors to execute applications, manage user sessions, and render content in real-time.
	16GB	
	8GB	
	8GB	
Display	19INCH1920 x 1080 (FULLHD) 60Hz	Offers expanded screen real estate to view code, live browser preview.
	1920 x 1200 60Hz	

	2736 x 1824 60Hz	
	1920 x 1080	
Accessories	Logitech G102	Perform basic functionalities
	Zeus MK440	
	Huawei MateView SE	
	Razer Viper Mini	
	ADM Razer Acer	

Table 3.1: Hardware resources utilized for the development of Xlore U

Software

Name	Description	Purpose
Visual Studio Code	Powerful source code editor supports multiple programming languages.	Used primarily for development environment
Node.js	Runtime environment that enables JavaScript to operate on the server level.	Used for backend development and server-side processing.
MySQL	An open-source relational database management system. Utilizing structured query language to store, organize, and manage data efficiently.	Primarily utilized as the database management system that stores and manages system data.
HTML5	Language used to create and	To create and structure the

	structure web pages.	content of web pages.
CSS3	Used to define styles for the web pages.	To Defined the visual and layout of website.
JavaScript	Programming language primarily used to create dynamic and interactive content on websites, supporting both frontend and backend development.	Main programming language used for both frontend and backend development.
Web Browsers	Software applications display web-based applications and content over the internet.	For testing and accessing the system.

Table 3.2: Software resources utilized for the development of Xlore U

Requirements Analysis

The system requirement analysis is concerned with assisting senior high school students in determining the appropriate college program and institutions. The primary users of the proposed system are senior high school students preparing for the tertiary level. Secondary users may include college students who aspire to examine alternative academic paths. Educators, parents, and guidance counselors might also utilize the systems to further guide students in their decision-making process. The focal point of the system is the college program selection and exploration. Providing users with access to program information, institution specializations, and personalized recommendations based on interests, skills, and preferences. The system would be accessible through the web, allowing users to access the platform at their convenience with the presence of internet connection. The system is mainly utilized during the transition period of students from secondary education to tertiary education. Although, the system may also be utilized when users require guidance in choosing or changing their academic programs or institutions. At the present, students rely on outdated and fragmented methods when choosing college programs. Which may include scouring for information on multiple platforms, asking peers or family members, or through consulting guidance counselors. A process that is time-consuming, confusing, and may not ultimately align with the student's interests, strengths, and preferences. The system strives to bridge the gap the outdated methods have created. Through a centralized, accessible, and personalized platform streamlining program and institution exploitation and supporting informed decision-making.

Requirements Documentation

A. Functional Requirements

User Account Management

REQ001. Users shall be able to register created accounts

REQ002. Users shall be able to login using their created credentials

REQ003. Users shall be able to update their credentials.

Program and Institution Exploration

REQ004. Users shall be able to browse the list of available college programs and institutions.

REQ005. The system shall display program description, requirements, and possible career paths.

REQ006. Users shall be able to search for programs and institutions based on keywords

Institution Information

REQ007. The system shall display tertiary institutions and their program offerings.

REQ008. Users shall be able to view institutional details such as location, program offerings, and tuition fees.

Recommendation System

REQ009. The system shall collect user interests, skills, and preferences.

REQ010. The systems shall be able to produce program and institution recommendations based on collected information

REQ011. Users shall be able to view program and institution recommendations.

Management of Preferences

REQ012. Users shall be able to save their preferred programs and institutions

REQ013. Users will be capable of removing saved programs and institutions

REQ014. The system shall display saved programs and institutions in the user dashboard.

Filtration System

REQ015. Users shall be able to filter programs and institutions based on specializations and tuition.

REQ016. The system shall display filtered results based on selected criteria.

B. Non-Functional Requirements

Operational Requirements

REQ017. The system shall operate on web applications.

REQ018. The system shall require internet connectivity.

REQ019. The system shall store data in a centralized database.

Performance Requirements

REQ020. The system shall display search results in less than (3) seconds.

REQ021. The system shall produce program and institution recommendations in real time.

Security Requirements

REQ022. The system shall require authentication before accessing user accounts.

REQ023. The system shall protect user data from unauthorized access.

REQ024. The system shall encrypt user passwords.

Usability Requirements

REQ025. The system shall provide a clear and intuitive navigation structure for users.

REQ026. The system shall allow users to complete program and institution search and recommendation with minimal steps.

Cultural and Language Requirements

REQ027. The system shall utilize English as the primary language.

Design of System and Processes

1. Landing Page and Account Creation

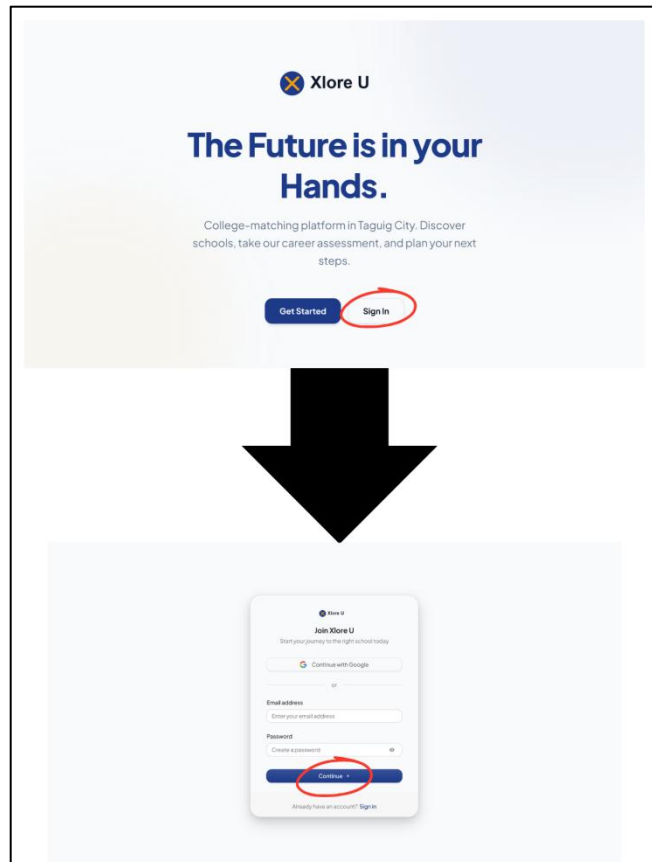


Figure 3.1: Landing page and Account creation flow

Upon launching the platform, students will be prompted to sign-in using their email accounts in order to provide personalization to their experience. For further authentication students are given a verification code through their emails before continuing to the platform.

2. Institution Comparison

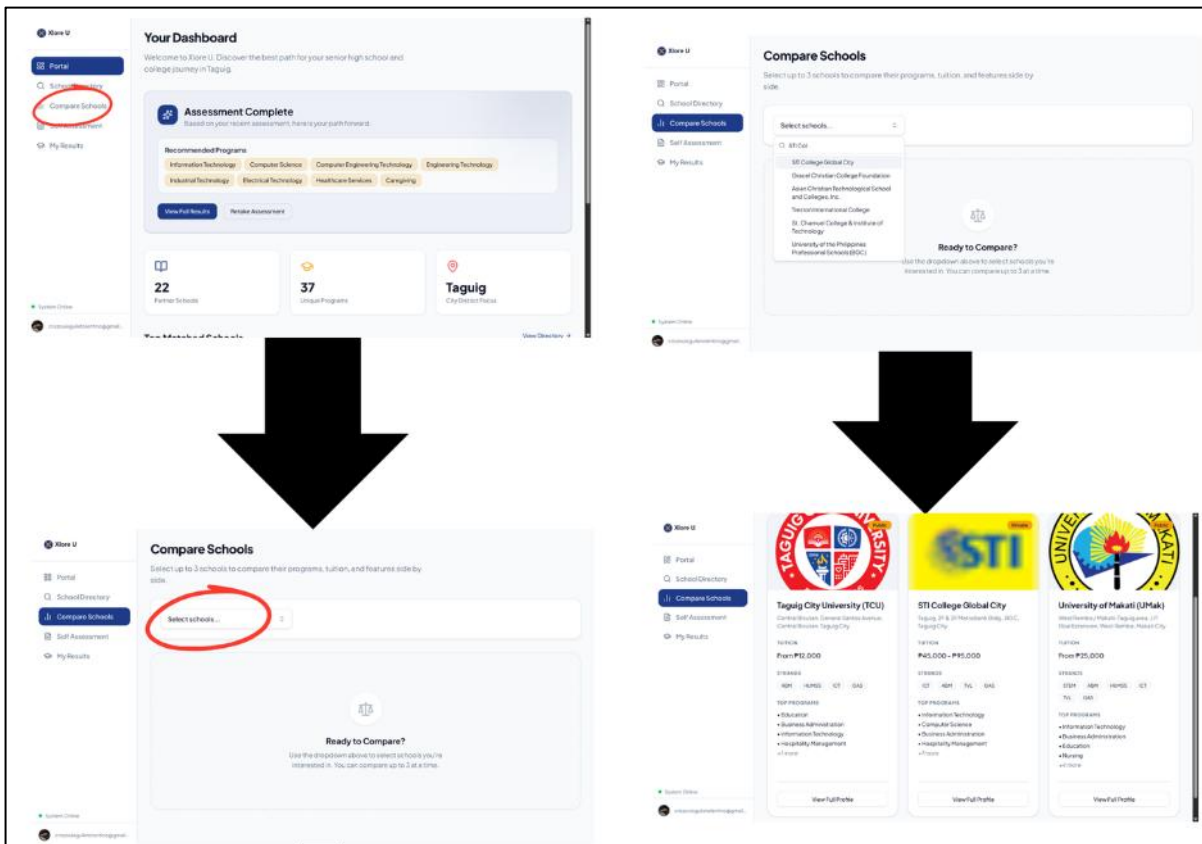


Figure 3.2: Institution Exploration flow

In terms of comparing institutions students are allowed to compare a maximum of three institutions. The platform provides the estimated tuition per semester, accreditation, location of the institution, and the programs offered by each institution. Users may also remove their prospective institutions based on their preferences.

3. School Directory

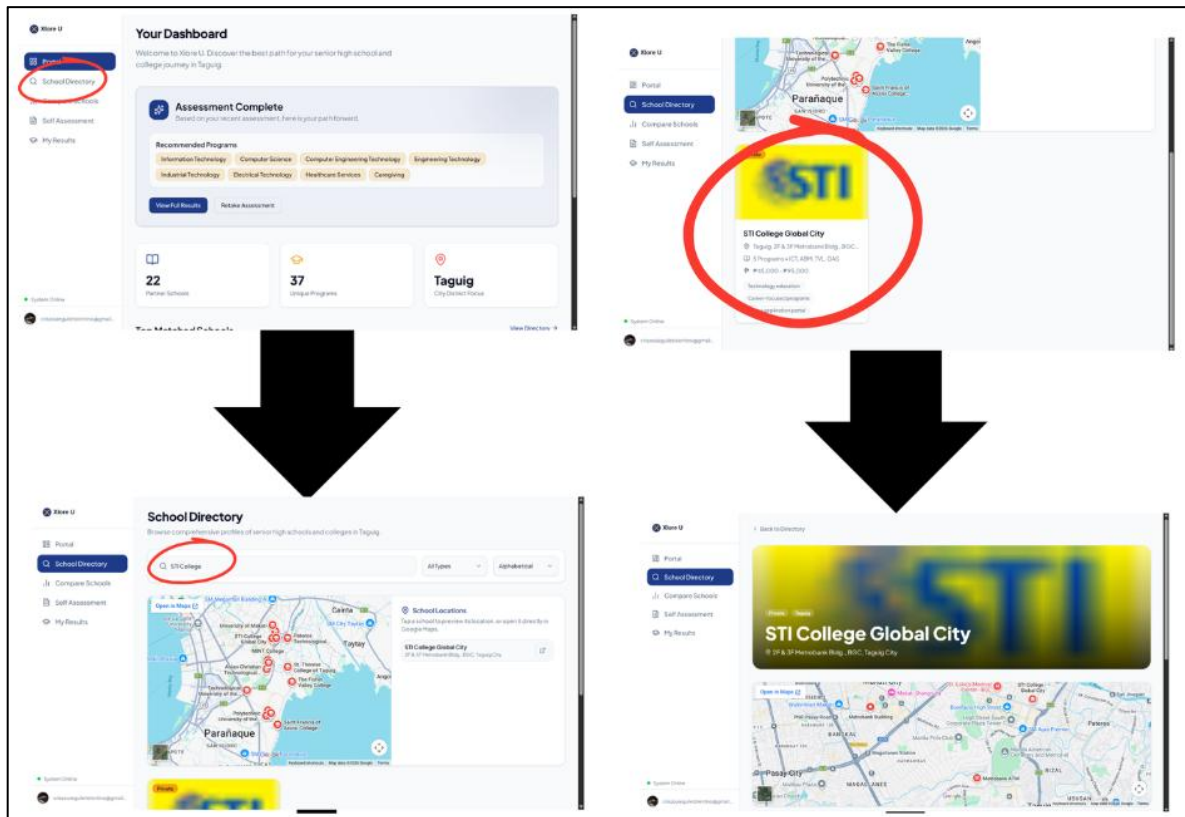


Figure 3.3: School Directory flow

Through pressing the School Directory button, students are directed to a map feature that locates the institution, along with relevant information about the institution.

4. Profile Assessment

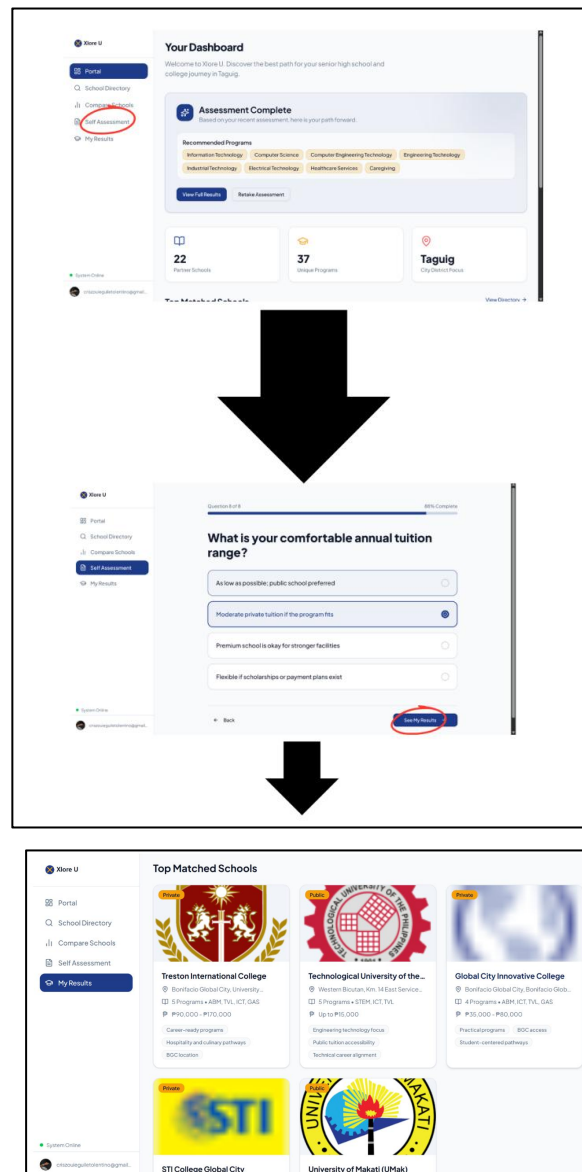


Figure 3.4: Profile Assessment flow

Users may also utilize the Start Assessment feature, which is a curated questionnaire aimed to identify the strengths and weaknesses of each student. After answering, the platform provides recommendations based on the results of the questionnaire.

5. School Recommendations

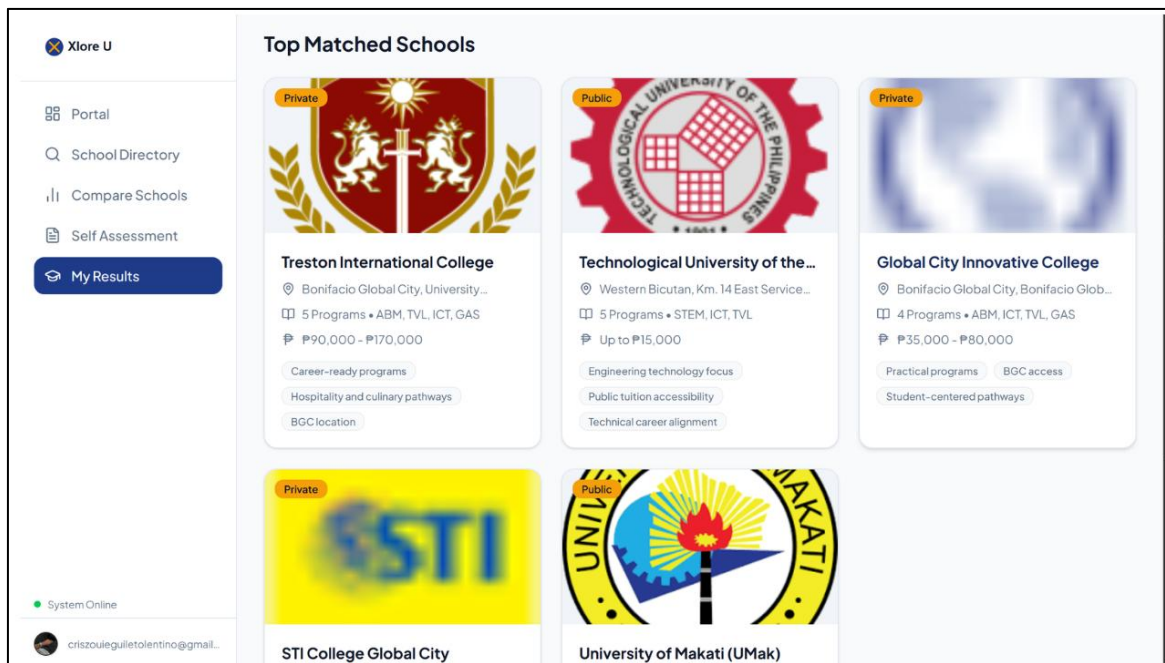


Figure 3.5: School Recommendations flow

Upon Accomplishing the questionnaire, Students are provided a curated list of recommended schools on their dashboard. Users are also provided a saved schools list based along with their assessment history and the option to retake their initial assessment.

6. Map Integration

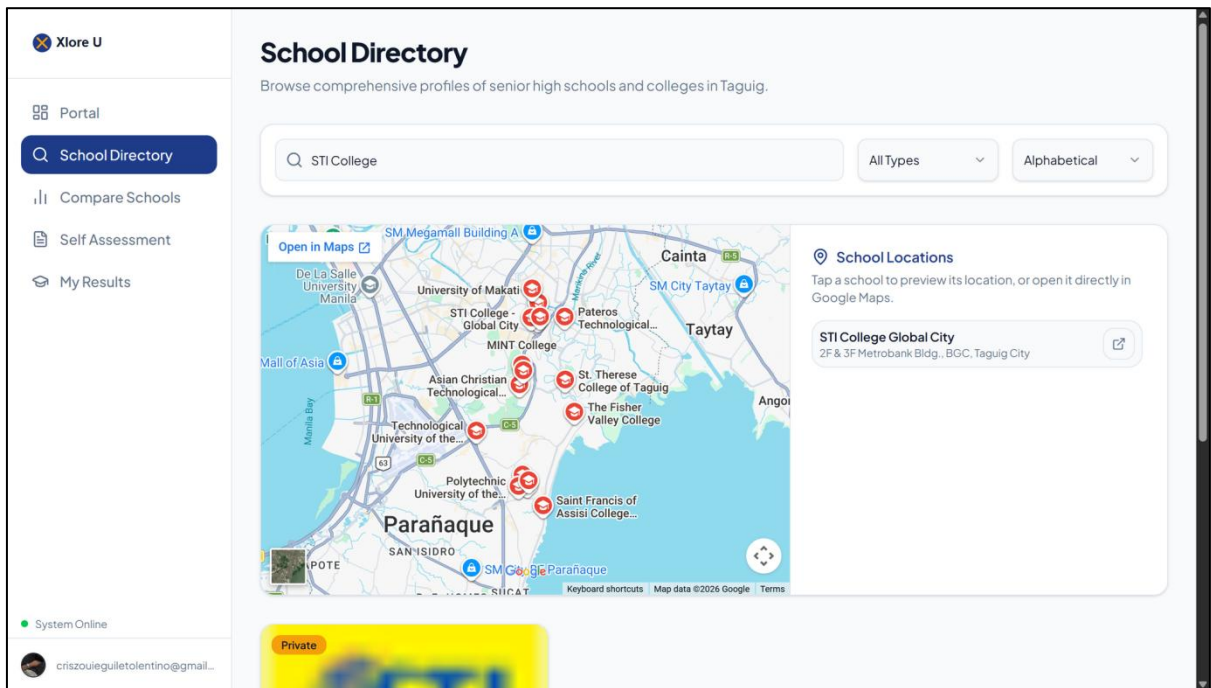


Figure 3.6: Map Integration Flow

Users may also locate their prospective institution with the Map Integration, users may click on the school directory button and preview the location of their prospective institution.

Database Design: Diagram of Entities and Relationships (ERD)

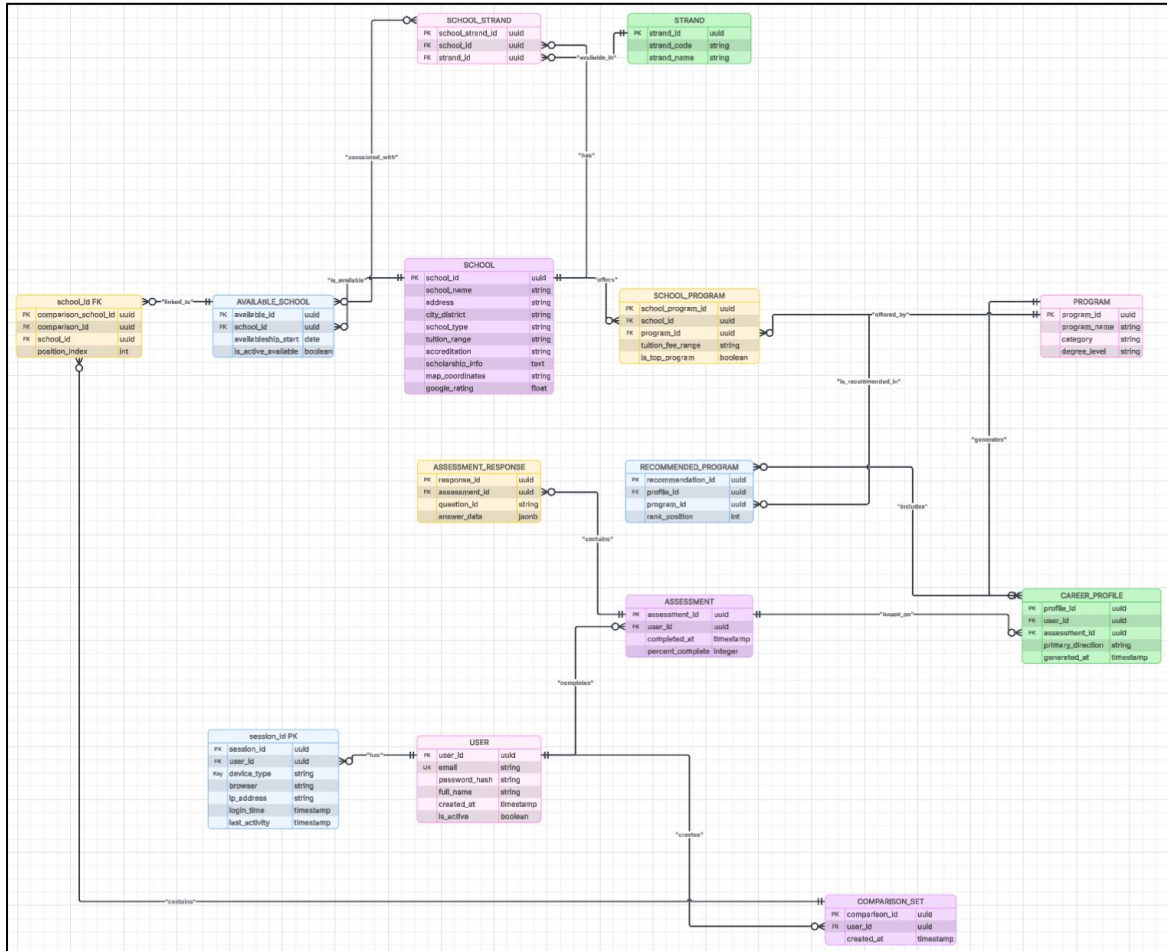


Figure 3.7: Xlore U, Diagram of Entities and Relationships Diagram

This Figure represents the fourteen entities that comprise the Xlore U system and their connections to one another. The primary function manages User accounts, along with authentication, session data, and activity. Followed by the Assessment which generates responses based on the results of user evaluations. Those results would then be integrated to the users profile, which recommends and suggests programs and schools based on the users specifications. The structure of the database organizes tertiary institutions through a centralized School entity. An entity that is linked to various program offerings via many-to-many relationships, to the school strand and school program entities. The system also includes a comparison feature, allowing users to evaluate multiple institutions at a time. The schema illustrates the coordinated ecosystem of Xlore U.

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APPENDICE

APPENDIX A. RESOURCE PERSONS

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of
<GIVEN NAME MI. FAMILY NAME>
<complete address>
<email address>
contact number either cellular phone or landline or both

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	month year	
Vocational/Technical	month year	
High School	month year	
Elementary	month year	

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).